



Munich Center for Machine Learning

Joint Model Importance via the Rashomon Set

Behavioural Comparison of Near-Equally-Good Models

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- In tabular ML, many models reach indistinguishable predictive performance.
- For a tolerance $\varepsilon > 0$, the *Rashomon set*

$$\text{RS}(\varepsilon) = \{ \hat{f}_{\mathcal{D},\lambda} : \mathcal{R}(\hat{f}_{\mathcal{D},\lambda}) \leq \mathcal{R}(\hat{f}_{\mathcal{D},\lambda^*}) \cdot (1 + \varepsilon), \lambda \in \Lambda \}$$

collects every model within ε of the best.

Notation: $\hat{f}_{\mathcal{D},\lambda}$ = model trained on data $\mathcal{D}$ with hyperparameter configuration $\lambda \in \Lambda$; λ^* = risk-minimising configuration; $\mathcal{R}(\cdot)$ = (empirical) risk functional.

- Members of $\text{RS}(\varepsilon)$ can disagree substantially on individual predictions and on feature importance.
- **Question.** How are these equally good models related to one another, and what does the collective tell us beyond the single best?

- ① For every $\hat{f}_{\mathcal{D},\lambda} \in \text{RS}(\varepsilon)$, compute the prediction vector $\hat{y}_\lambda$ on a held-out 2/3 validation split.
- ② Build the pairwise *behavioural distance* matrix D with entries

$$d_{ij} = \|\hat{y}_i - \hat{y}_j\|_p = \left(\sum_{o=1}^{n_{\text{val}}} |\hat{y}_{i,o} - \hat{y}_{j,o}|^p \right)^{1/p},$$

where i, j index models in $\text{RS}(\varepsilon)$ and o runs over the n_{val} validation observations (see backup slide).

- ③ View D three ways:
 - ▶ hierarchical clustering (`hclust`)
 - ▶ classical MDS (`cmdscale`, embedding dimension $k = 2$)
 - ▶ partitioning around medoids (`cluster::pam`)

Predictions are class probabilities for classification, $\hat{f}(x)$ for regression. Distances computed for both Euclidean ($p = 2$) and Manhattan ($p = 1$); results shown are Euclidean (Manhattan tells the same story).

Rashomon tolerance $\varepsilon = 1\%$ of the minimum risk.

Five learner families considered for every task: `tree`, `glmnet`, `xgb`, `nnet`, `svm`.

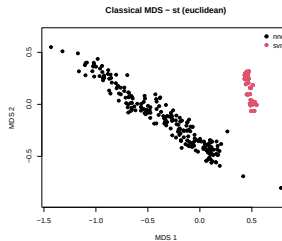
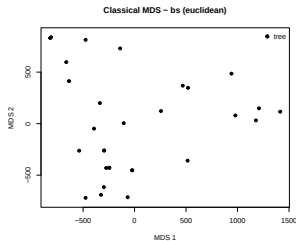
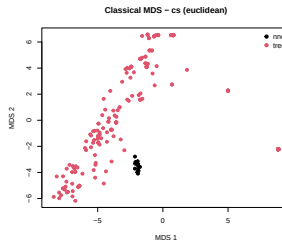
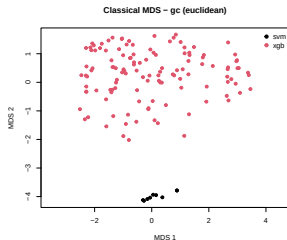
Key	Task	Type
gc	German Credit	classification
cs	COMPAS	classification
bs	Bike Sharing	regression
st	Synthetic	regression

Who survives the Rashomon cut?

Task	Surviving families (n models)
gc	svm(9), xgb(125)
cs	nnet(15), tree(261)
bs	tree(30)
st	nnet(216), svm(160)

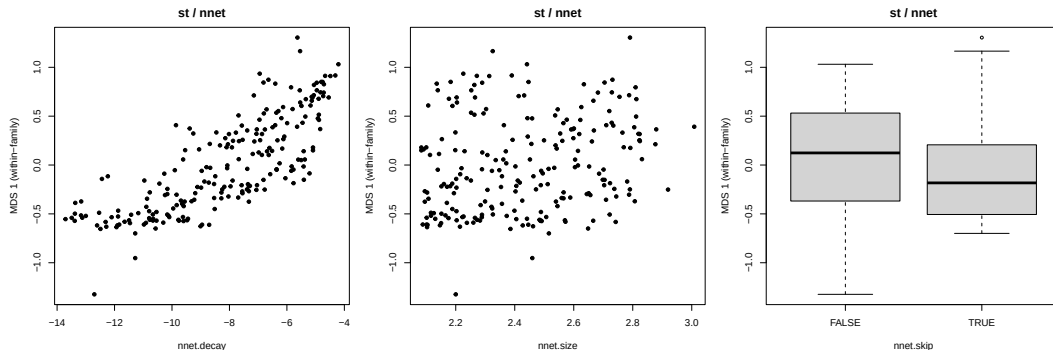
First finding. At $\varepsilon = 1\%$ only one or two learner families survive per task — *learner choice* is the first dimension of disagreement inside $RS(\varepsilon)$.

MDS view — all four tasks



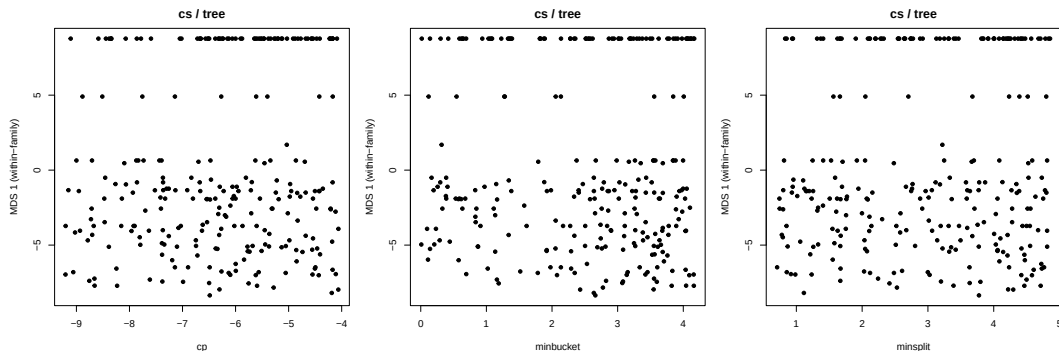
- Wherever two families coexist, behavioural distance **separates them cleanly** — the dominant axis is family identity, not within-family variation.
- Within-family variance is *asymmetric*: `svm` and `nnet` form tight clusters; `tree` and `xgb` scatter widely despite fewer tunable hyperparameters.
- Several families trace out a **1D-manifold** in the MDS plane (e.g. `st/nnet`, `cs/tree`) — next two slides.

Within-family structure I — st / nnet



- Each panel: 1D within-family MDS coordinate vs. one hyperparameter.
- The within-family manifold **is the decay axis** — near-monotone mapping.
- size and skip have no effect on behavioural position.

Within-family structure II — cs / tree



- Within-family MDS coordinate vs. `cp`, `minbucket`, `minsplit`.
- Trees on COMPAS are **bimodal**: a tight cluster at $\text{MDS } 1 \approx 8$ and a diffuse population below zero.
- *No single tuned hyperparameter* separates the two modes. **Speculation**: a discrete behavioural split driven by structure (e.g. root split, depth) that hyperparameter tuning does not reach — to be confirmed.

- **Joint model importance is a two-level question:**
 - ① Which learner family does the model come from?
 - ② Where on the family's behavioural manifold does it sit?
- Across every task observed, learner choice is the first axis of disagreement — not hyperparameter setting.
- Within-family structure can be smooth and HP-aligned ($\text{st}/\text{nnet} \sim \text{decay}$) or discrete and HP-orthogonal (cs/tree ; *causal mechanism still speculative*).

- **ε sensitivity.** How does family composition shift as the Rashomon tolerance opens?
- **Feature-importance distance.** Compare behavioural distance with a distance built from VIC — do the clusters agree?
- **Choose the number of clusters k for PAM via silhouette,** then read out the medoids as “representative” members of $RS(\varepsilon)$.
- **Disagreement map.** Where in feature space do Rashomon-set models disagree the most?
- **Discrete tree mode on cs.** Is the bimodality driven by structural differences (root split, depth) that hyperparameters do not capture?

Thank you.

Questions?

Stack the Rashomon-set prediction vectors row-wise into a prediction matrix:

$$\mathbf{P} = \begin{pmatrix} \hat{y}_{1,1} & \hat{y}_{1,2} & \cdots & \hat{y}_{1,n_{\text{val}}} \\ \hat{y}_{2,1} & \hat{y}_{2,2} & \cdots & \hat{y}_{2,n_{\text{val}}} \\ \vdots & \vdots & & \vdots \\ \hat{y}_{n,1} & \hat{y}_{n,2} & \cdots & \hat{y}_{n,n_{\text{val}}} \end{pmatrix} \in \mathbb{R}^{n \times n_{\text{val}}}$$

with one row per model in $\text{RS}(\varepsilon)$ (here $n = |\text{RS}(\varepsilon)|$) and one column per held-out validation observation (here n_{val}).

Row-wise L_p distances collapse the observation axis and yield the symmetric matrix

$$\mathbf{D} \in \mathbb{R}^{n \times n}, \quad d_{ij} = \|\hat{y}_i - \hat{y}_j\|_p.$$

Rows / columns of $\mathbf{D}$ now index *models*, not observations. Hence **one MDS point = one model**; the cloud sits in $\mathbb{R}^n$ and is visualised in $\mathbb{R}^2$ via `cmdscale`.